

Part XII. Compare the Documented and Artificial Mutations

	WT Sequence	Documented mutation	Artificial mutation
CDS Length	2,763 bp	2,763 bp	2,763 bp
Protein Length	920 aa	920 aa	920 aa
Mutation Type	None	Missense	Missense
Reading frame changed	Frame 1	Unchanged	Unchanged
Premature stop codon appeared	No	No	No
Amino acids affected	None	Ala766 → Thr766	Glu2 → Ala2
Expected functional consequence	Normal androgen receptor function.	It may alter the androgen receptor function and androgen signaling.	It may alter receptor function because of the amino-acid substitution.

The wild-type AR sequence contains 2,763 bp and produces a 920-amino-acid protein. The documented mutation changes alanine 766 to threonine, while the artificial mutation changes glutamic acid 2 to alanine. Both are missense mutations that leave the reading frame and protein length unchanged but may have different effects on AR function.